Sequences, Series & Complex Numbers

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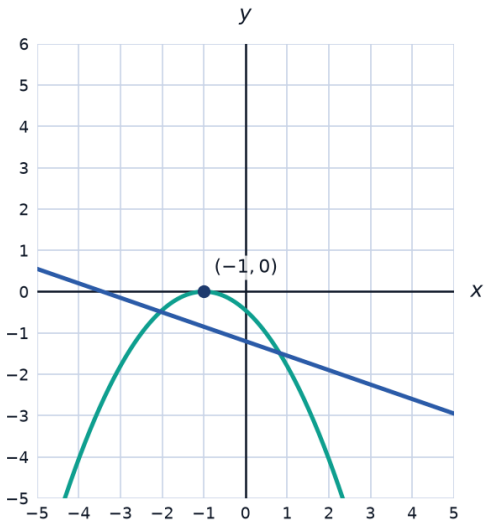
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences**

Find the 12th term of the arithmetic sequence $5, 9, 13, \ldots$

2 **[Foundational]** • **Target: Arithmetic sequences**

The 4th term of an arithmetic sequence is 18, and the 9th term is 38. Find the first term and the common difference.



3 **[Foundational]** • **Target: Arithmetic sequences**

Find the sum of the first 20 terms of the arithmetic sequence $7, 11, 15, \ldots$

4 **[Foundational]** • **Target: Arithmetic sequences**

Find the number of terms in the arithmetic sequence $6, 10, 14, \ldots, 150$.

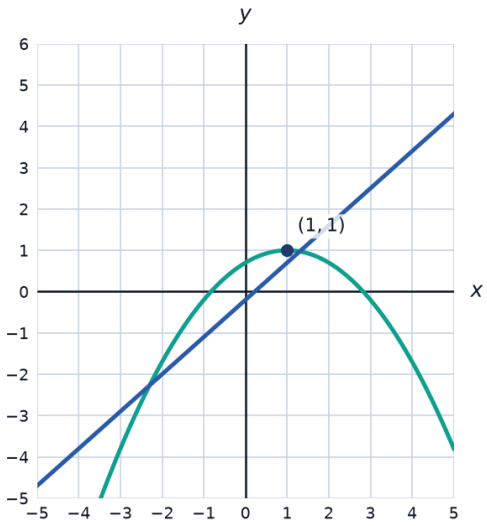
5 **[Foundational]** • **Target: Geometric sequences**

Find the 7th term of the geometric sequence $3, 6, 12, \ldots$

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6 [Foundational] • Target: Geometric sequences

In a geometric sequence with all positive terms, the 3rd term is 20 and the 5th term is 80. Find r and a_1.



7 [Foundational] • Target: Geometric sequences

Given that $x - 1$, $x + 1$, and $3x - 1$ are consecutive terms of a geometric sequence, find the possible value(s) of x.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

Find the sum of the first 6 terms of the geometric sequence $5, 15, 45, \ldots$

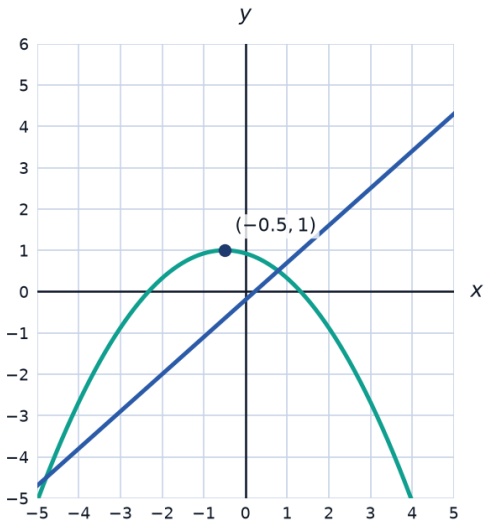
9 [Intermediate] • Target: Series and sigma notation

Write the series $3 + 7 + 11 + \cdots + 39$ in sigma notation, then evaluate it.

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10 [Intermediate] • Target: Geometric series

Find the sum to infinity of $16 - 8 + 4 - 2 + \cdots$



11 [Intermediate] • Target: Geometric series

Express the repeating decimal $0.4\overline{5}$ (that is, $0.45555\ldots$) as a geometric series and find its exact fractional value.

12 [Intermediate] • Target: Complex number arithmetic

Simplify i^{23}, using $i^2 = -1$.

13 [Intermediate] • Target: Complex conjugates and division

Find the conjugate of $z = 6 + 7i$, and compute $z\bar{z}$.

14 [Intermediate] • Target: Complex conjugates and division

Simplify $\frac{4+3i}{1-2i}$ by multiplying by the conjugate of the denominator.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

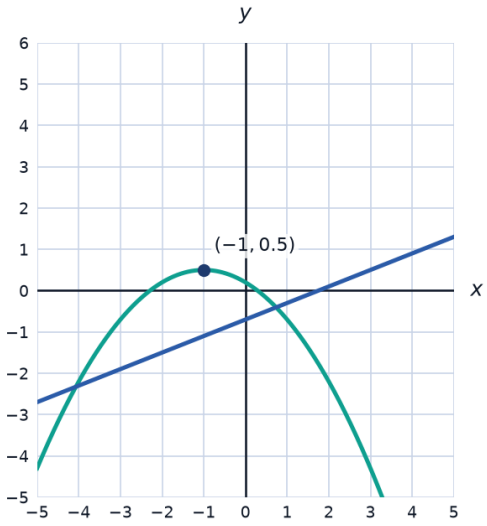
15 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve $x^2 + 4x + 13 = 0$, giving your answer in the form $a + bi$.

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16 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 5.01 Trigonometry

One root of a quadratic equation with real coefficients is $2 - 5i$. Find the other root, and write the equation in standard form.



17 [Challenging] • Target: A well-designed word problem: a bouncing ball • Cross-Lesson: 6.01 Circles & Coordinate Geometry

This question has three parts. A ball is dropped from a height of 20 meters. Each time it bounces, it rises to 60% of its previous height. (a) Find the height of the ball's rebound after its 4th bounce. (b) Write an expression for the total vertical distance traveled by the ball as it bounces infinitely many times, and evaluate it. (c) After which bounce does the rebound height first fall below 1 meter?

18 [Challenging] • Target: A well-designed word problem: a bouncing ball • Cross-Lesson: 7.01 Solid Geometry & Similarity

Let $z = \frac{5+2i}{2-3i} + i^{43}$. (a) Express z in standard form $a + bi$. (b) Compute the modulus $|z|$.

19 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 8.01 Statistics & Probability

The sum of the first n terms of an arithmetic sequence is given by $S_n = 2n^2 + 3n$. Find the 10th term a_{10}.

20 [Challenging] • Target: Geometric sequences • Cross-Lesson: 1.01 Advanced Algebra

An infinite geometric series has first term $a_1 = 18$ and sum to infinity $S_\infty = 24$. (a) Find the common ratio r. (b) Calculate the third term a_3.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Arithmetic sequences

Find the 3rd term of the geometric sequence $2, 6, 18, \ldots$

a. 16

b. -18

c. 18

d. 20

22 [Foundational] • Target: Geometric sequences

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$

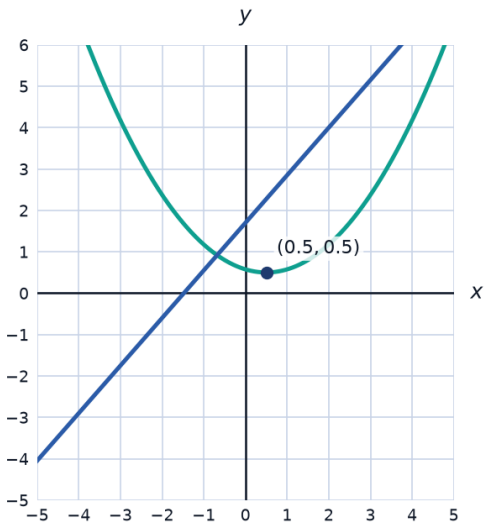
b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

23 [Foundational] • Target: Series and sigma notation

An arithmetic sequence has its n-th term given by $a_n = 3n + k$, where k is a constant. If the sum of the first 5 terms of this sequence is 65, find the value of k.



a. 1

b. 2

c. 3

d. 4

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24 [Foundational] • Target: Geometric series

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fifth term of this sequence?

a. 17

b. 14

c. 20

d. 23

25 [Foundational] • Target: Complex number arithmetic

A geometric sequence has its first term $a_1 = 3$ and a common ratio $r = 2$. What is the third term, a_3, of this sequence?

a. 6

b. 9

c. 12

d. 15

26 [Foundational] • Target: Complex conjugates and division

An arithmetic sequence has $a_3 = 0$ and $a_5 = -4$. Find S_{12}, the sum of the first 12 terms.

a. -92

b. -84

c. -85

d. -79

27 [Foundational] • Target: Modulus of complex numbers

In an arithmetic sequence, $a_4 = -1$ and $a_8 = -9$. Find the common difference d.

a. -2

b. -3

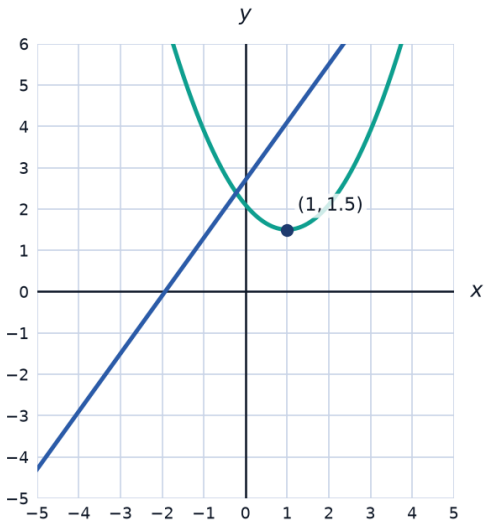
c. -6

d. 6

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28 [Foundational] • Target: Quadratic equations with complex roots

Find the sum to infinity of the geometric series with $a_1 = 4$ and $r = 2/3$.



a. 6

b. 12

c. 19

d. 15

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a bouncing ball

An arithmetic sequence has first term $a_1 = -1$ and common difference $d = 4$. Find a_6.

a. 21

b. 15

c. 26

d. 19

30 [Intermediate] • Target: Arithmetic sequences

A geometric sequence has first term $a_1 = 2$ and common ratio $r = 1/2$. Find a_5.

a. -3/8

b. 1/8

c. 17/8

d. 3/8

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31 [Intermediate] • Target: Geometric sequences

A sequence is defined by the recursive relation $a_{n+1} = 2a_n$ with the first term $a_1 = 5$. What is the third term of this sequence?

a. 10

b. 20

c. 15

d. 25

32 [Intermediate] • Target: Series and sigma notation

Consider the quadratic equation $z^2 - (3 + 2i)z + (1 + 3i) = 0$. Let the roots of this equation be α and β. If the complex number w is defined such that $w = \frac{\alpha^2 + \beta^2}{2}$, find the modulus of w, $|w|$.

a. $\sqrt{5}$

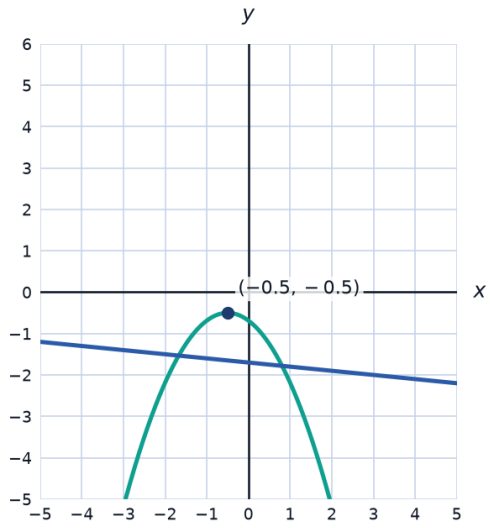
b. $\frac{\sqrt{5}}{2}$

c. $\sqrt{29}$

d. $\frac{\sqrt{29}}{2}$

33 [Intermediate] • Target: Geometric series

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?



a. $\frac{5}{4}$

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

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34 [Intermediate] • Target: Complex number arithmetic

Consider the quadratic equation $z^2 - (3 + i)z + (2 + 2i) = 0$, where z is a complex number. Let the roots of this equation be z_1 and z_2. Determine the value of $|z_1 - z_2|^2$.

a. 1

b. 2

c. 4

d. 5

35 [Intermediate] • Target: Complex conjugates and division

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.

a. 6

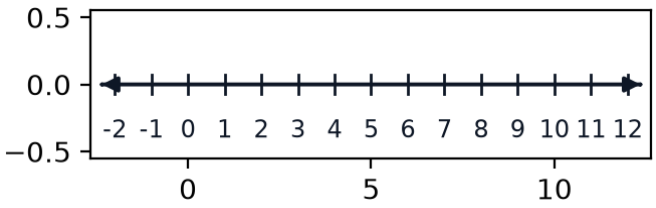
b. 8

c. 10

d. 12

36 [Intermediate] • Target: Modulus of complex numbers

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?



a. 3

b. -3

c. 12

d. 6

37 [Intermediate] • Target: Quadratic equations with complex roots

Let m be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots z_1 and z_2. In the complex plane, the origin $O(0)$ and the points representing z_1 and z_2 form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

a. $15\sqrt{3}$

b. $12\sqrt{3}$

c. $\frac{15\sqrt{3}}{2}$

d. $9\sqrt{3}$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a bouncing ball • Cross-Lesson: 7.01 Solid Geometry & Similarity

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

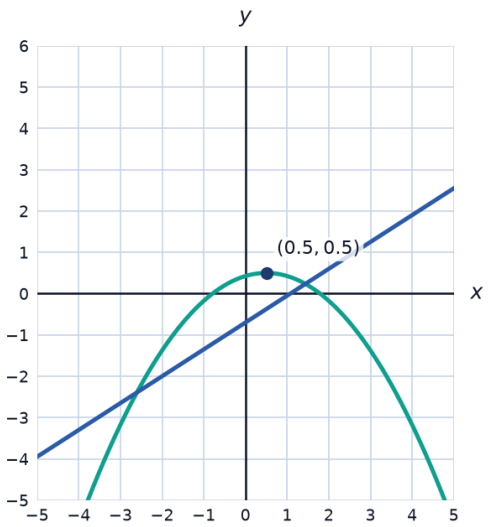
b. 72

c. 96

d. 192

39 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 8.01 Statistics & Probability

Consider the quadratic equation $2x^2 - (k-1)x + (k+1) = 0$ where k is a real constant. The equation has two non-real complex conjugate roots. If the imaginary parts of the roots have a magnitude of $\frac{\sqrt{7}}{2}$, what is the greatest possible value of k?



a. 7

b. 3

c. 10

d. 5

40 [Challenging] • Target: Geometric sequences • Cross-Lesson: 1.01 Advanced Algebra

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of this function?

a. $x = 2$ and $x = 4$

b. $x = -2$ and $x = -4$

c. $x = 6$ and $x = -8$

d. $x = 1$ and $x = 8$

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41 [Challenging] • Target: Series and sigma notation • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

42 [Challenging] • Target: Geometric series • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?

a. $x = 3$ and $x = 5$

b. $x = -3$ and $x = -5$

c. $x = 3$ and $x = -5$

d. $x = -3$ and $x = 5$

43 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 5.01 Trigonometry

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. 5

b. 7

c. 25

d. 16

44 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. 5

b. 7

c. 12

d. 25

45 [Challenging] • Target: Modulus of complex numbers • Cross-Lesson: 7.01 Solid Geometry & Similarity

A quadratic with roots $r_1 = 2$ and $r_2 = 5$ has equation:

a. $x^2 - 3x + 10 = 0$

b. $x^2 - 10x + 7 = 0$

c. $x^2 - 7x + 10 = 0$

d. $x^2 + 7x + 10 = 0$

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